

ElasticUMA: Reload-Correct OS-Managed Expert Caching for Apple Silicon

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Code and artifact: <https://github.com/Naresh084/elasticuma>

Abstract

Storage-backed MoE inference on Apple Silicon has a shared-memory problem: one physical pool serves the CPU, GPU, filesystem cache, applications, compressor, and swap. For storage-backed mixture-of-experts (MoE) inference, a large explicit expert cache can therefore raise its local hit rate while increasing system pressure and reducing end-to-end throughput. This paper introduces ElasticUMA, a reload-correct expert cache that separates logical capacity from physical entitlement. Each expert slot is a Metal-owned shared buffer. The runtime keeps a small hot set nonvolatile and marks cold slots volatile only at synchronized prefill-layer and decode-token boundaries. Before treating a volatile mapping as a hit, it relocks the buffer. If Metal reports that the contents are empty, ElasticUMA invalidates the mapping and reloads the exact expert bytes before GPU binding.

We evaluate the committed mechanism against untouched fixed-16 and fixed-96 upstream policies on an M1 Max with 32 GiB unified memory. Under a bounded 4 GiB co-tenant, Qwen3.6-35B-A3B Q4 gains 7.25% and 13.67% paired-median decode throughput over the two baselines, while reducing fixed-96 peak physical footprint by 31.64%. Without synthetic pressure, the gains are 16.91% and 17.59%, with a 31.65% footprint reduction. A second architecture, Gemma 4 26B-A4B Q4, gains 45.30% and 26.01% and reduces fixed-96 footprint by 35.26%. All 45 positive measured rows pass admission with within-protocol output equality and observed empty-slot recovery. The contribution is a two-architecture mechanism on one Mac that lets a large logical cache coexist with OS-wide reclamation; it does not establish cross-generation, energy, concurrency, or independent-reproduction claims.

Keywords: Apple Silicon, unified memory, mixture-of-experts inference, Metal, purgeable memory, expert caching.

1 Introduction

Open-weight mixture-of-experts (MoE) models decouple total model capacity from the computation performed for each token. A router selects a small subset of experts, so a model with tens or hundreds of billions of parameters may execute only a few billion active parameters at a time. That sparsity makes local serving possible, but it does not make the inactive weights disappear. A local runtime must still place the complete expert pool and move requested experts into execution memory. It must also decide which cached state should survive when the machine becomes busy.

FreeToken addresses this problem on heterogeneous CUDA systems by combining a host-resident expert pool, a GPU expert cache, double-buffered prefill, and a bandwidth-calibrated choice between CPU execution and PCIe transfer [1]. KTransformers similarly demonstrates that CPU-GPU hybrid execution can make large MoE checkpoints practical on commodity machines [2]. These systems exploit a resource model with distinct host and device memories. Apple Silicon changes that model. The CPU and GPU share physical memory, and Metal shared buffers do not cross a PCIe boundary. Expert slots, immutable file-backed weights, prompt state, the filesystem cache, the display stack, desktop applications, compression, and swap all compete in the same pool.

This shared pool creates a failure mode that is invisible to cache-local metrics. We first characterize whether cache hit rate tracks end-to-end service performance. Increasing explicit expert capacity can raise the hit rate yet evict useful file-backed pages, increase compression, and reduce decode speed. Our preregistered fixed-cache sweep demonstrates that tension. Increasing Qwen from 16 to 96 expert slots per layer raises median hit rate from 48.98% to 90.13%, while median decode throughput falls from 11.330 to 10.120 token/s. Peak compressor growth also increases by roughly 5 GiB. The paired slowdown is 10.54%, below the frozen 15% threshold required to authorize our original cache-size controller. We retain that negative result and abandon the controller rather than lower its gate.

The fixed-cache curve suggests a different mechanism. The runtime benefits from the logical identity and reuse information of a large cache, but it need not claim permanent physical residency for every cold slot. Apple exposes

purgeable Metal resources for exactly this cache-like relationship: an application may mark a resource volatile, the operating system may reclaim its contents, and the application must relock and recreate or reload the resource before access [3, 4]. Applying that contract to model weights is not automatic. A cache mapping can outlive the physical contents it names; a GPU command may still reference a resource when a state transition is attempted; and externally allocated memory wrapped by Metal may remain physically owned even when the wrapper is marked purgeable.

We present ElasticUMA to resolve those correctness and integration problems. Each expert slot is allocated through Metal and retains a stable CPU pointer for positional reads. The cache tracks each slot as nonvolatile, volatile, or empty. At safe boundaries it keeps the hottest slots nonvolatile and offers the cold remainder to macOS. A future logical hit is provisional until a nonvolatile relock confirms that its contents survived. An empty return clears identity and recency state and enters the existing exact miss path. The GPU sees either validated cached bytes or a fully reloaded expert; discarded contents cannot reach a command encoder.

This paper makes four contributions:

- It characterizes the mismatch between expert-cache hit rate and end-to-end service performance in one Apple unified-memory system, including a preregistered negative result that stops an earlier controller claim.
- It presents a reload-correct state machine for Metal-owned expert slots, with explicit validity, publication, and GPU-synchronization invariants.
- It integrates selective purgeability at the actual prefill-layer and decode-token boundaries, avoiding the full physical prefill peak that defeats a post-prefill-only policy.
- It evaluates the committed implementation against untouched small- and large-cache upstream binaries on Qwen3.6 and Gemma 4, with balanced held-out repetitions, physical-footprint telemetry, exact-recovery receipts, negative alternatives, immutable artifacts, and explicit claim limits.

The strongest supported conclusion is intentionally scoped. On the measured M1 Max workloads, ElasticUMA improves the throughput-footprint frontier: it dominates fixed-96 with higher throughput and lower footprint, while providing more throughput and locality than fixed-16 at a higher footprint. One host and two architectures do not establish an all-Mac result. Cross-chip reproduction, energy measurement, longer contexts, concurrency, and independent operation remain required.

2 Background and Motivation

2.1 Storage-backed routed-expert inference

An MoE layer contains a router, a shared or dense path, and a pool of routed experts. For each token the router selects a small top- k set. A bounded-memory runtime can keep non-expert tensors resident while reading selected expert tensors from storage into a fixed number of execution slots. MoE-Infinity shows that request-level sparsity and expert reuse support prefetching and caching on personal machines [5]. MawForge demonstrates a related storage-backed design on constrained unified-memory systems and finds that maximum cache capacity is not itself an optimal policy [6].

The executor used in this study descends from slipstream [7]. It stores immutable layer files in a packed `.gturbo` directory, maintains a bounded per-layer routed-expert cache, loads misses with positional reads, and executes native Metal kernels. The upstream fixed policy owns every allocated slot until the model unloads. This provides a strong controlled baseline because the model, quantization, kernels, tokenizer, storage layout, and request path remain constant while cache residency changes.

2.2 Why Apple unified memory is not free capacity

Unified addressing removes explicit host-to-device copies, but it does not remove physical scarcity. The same DRAM services CPU and GPU demand, file-backed pages, anonymous allocations, and the compressor. A process’s resident set size also becomes an unreliable measure of its current physical claim: a purgeable page may remain mapped and counted in RSS while the operating system is free to discard its contents. We therefore use `proc_pid_rusage().ri_phys_footprint` as the primary process metric and report RSS only as secondary context.

Apple documents `MTLPurgeableState.volatile` as a cache state whose contents may be discarded, and `setPurgeableState` returns the prior state when an application relocks a resource [3]. This creates two distinct notions of capacity:

- **Logical capacity** is the number of expert identities and stable Metal resources represented by the cache.

- **Physical entitlement** is the subset of pages the application currently insists must remain valid.

A fixed cache equates the two. ElasticUMA deliberately separates them.

2.3 Prefill and decode expose different residency shapes

Prompt prefill processes many tokens in parallel and often activates a broad union of experts. Decode processes one new token at a time and exhibits stronger short-range route reuse. FreeToken treats these phases differently through full-layer streaming in prefill and semantic-aware caching in decode [1]. The phase distinction is also necessary on Apple UMA, but for a different reason. If cold slots become volatile only after all prefill completes, every layer can materialize its large cache first. The host has already paid the peak physical cost before the policy runs. ElasticUMA must offer cold pages to the OS immediately after each layer’s routed GPU work finishes.

2.4 The rejected cache-size controller

Our initial hypothesis was that a live pressure signal could select among fixed cache sizes. We froze a 15% throughput-gap gate before measuring Qwen. The fixed sweep found a stable 10.54% paired slowdown from 16 to 96 slots, with five of five pairs in the same direction, but it did not reach the gate. This result has two implications. First, the pressure tradeoff is real but not large enough to justify a controller built around ordinary cache-size switching. Second, changing the threshold after observing the data would turn a falsification gate into post-hoc selection. We instead test a new mechanism with new prompts, new experiment identifiers, and separately frozen gates.

Gate 1 fixed expert-cache sweep on M1 Max

Qwen3.6-35B-A3B Q4 · AC power · n=5 admitted runs per arm · identical outputs

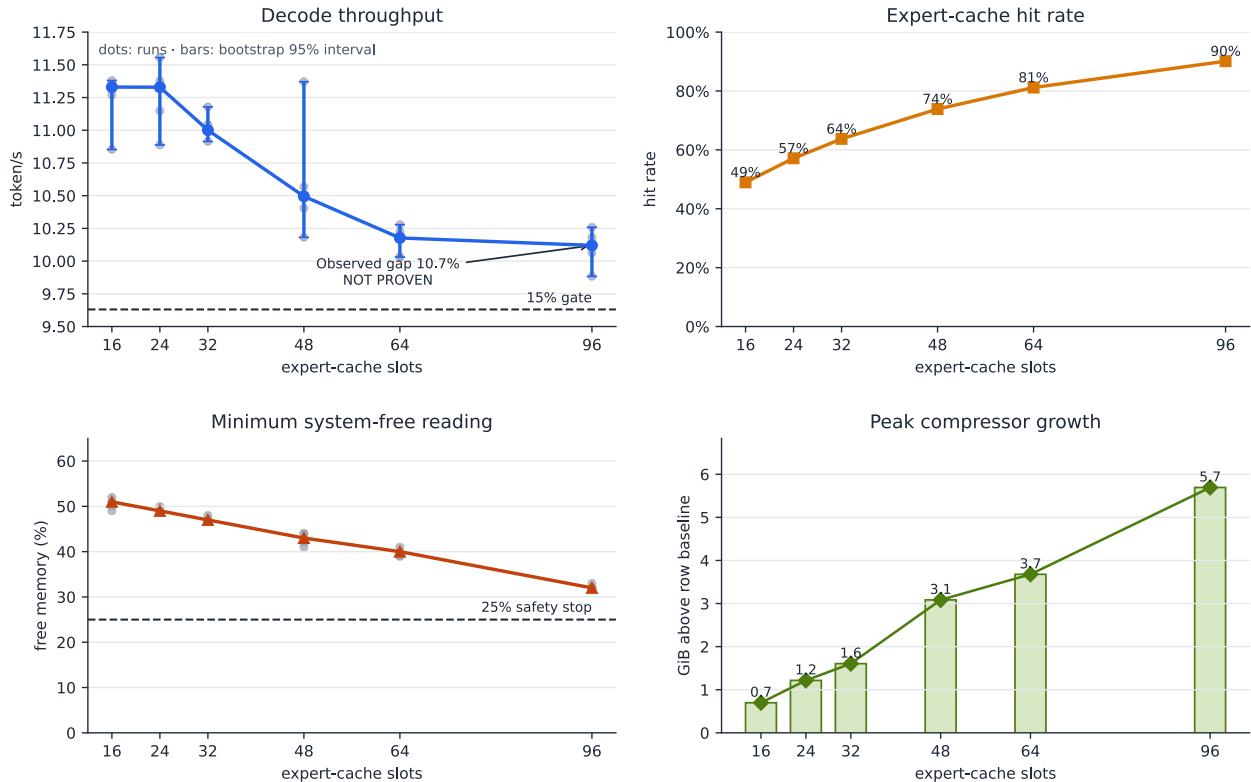


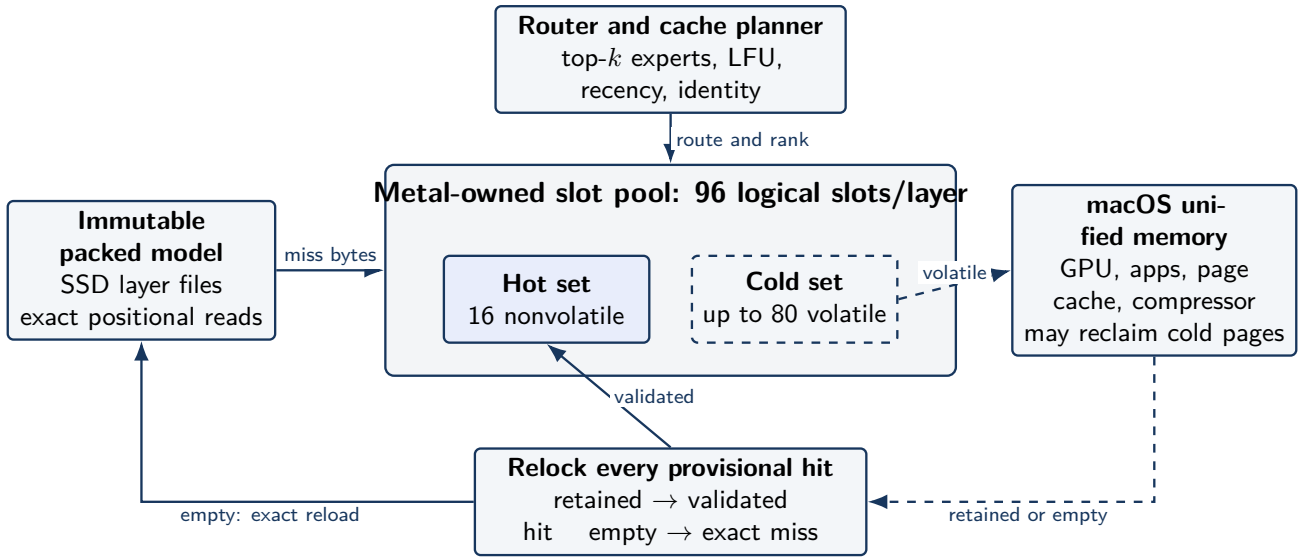
Figure 1: Fixed-cache characterization on Qwen3.6-35B-A3B Q4. More slots monotonically raise hit rate but reduce decode throughput and free-memory headroom while increasing compressor growth. The observed 16-to-96 slowdown misses the preregistered 15% controller-authorization gate. Bars and points summarize five admitted measured repetitions per arm.

3 ElasticUMA Design

3.1 Design goals

ElasticUMA is guided by four goals.

1. **Exactness.** Reclamation may change latency and I/O, but never routing, weights, generated semantics, or the bytes observed by a GPU kernel.
2. **Stable resources.** Existing Metal bindings and CPU-side positional reads should not require per-miss resource allocation.
3. **Timely physical elasticity.** Cold pages must become reclaimable before later prefill layers create the global physical peak.
4. **Baseline isolation.** Fixed residency must bypass the new state machine, preserving an untouched behavioral comparison.



Safe points change physical residency; the runtime retains identity and exactness.

Figure 2: ElasticUMA separates logical cache identity from physical residency. The packed model is immutable. Hot Metal-owned slots remain nonvolatile; cold slots become volatile after GPU-safe boundaries. Every provisional hit is relocked. If macOS reports an empty prior state, the mapping is invalidated and exact bytes are reloaded before GPU use.

3.2 Metal-owned per-expert slots

The upstream executor allocates anonymous CPU memory and wraps it in `bytesNoCopy` Metal buffers. An isolated probe showed that making such a wrapper purgeable does not surrender the external allocation: Metal does not own the pages. ElasticUMA instead allocates each slot with `MTLDevice.makeBuffer(storageModeShared)` and uses the stable `buffer.contents()` pointer as the destination of the existing exact `pread` path. The Metal object and CPU pointer survive transitions among nonvolatile, volatile, empty, and refilled states.

This choice is essential. Reallocating a resource on every empty recovery would change binding identity, increase synchronization work, and complicate cache publication. Stable Metal ownership lets physical contents disappear while the runtime retains the slot object, its index, and its synchronization discipline.

3.3 Residency and identity state

For each slot s , the cache stores a Metal buffer b_s , an optional expert identity e_s , LFU count f_s , recency timestamp r_s , and residency state q_s . The state q_s is one of:

State	Physical contract	Valid use
Nonvolatile	macOS must preserve contents	May be bound after normal cache planning
Volatile	macOS may discard contents	Must be relocked before a hit is accepted
Empty	Prior contents are invalid	Identity is cleared; exact reload is mandatory

Logical identity and physical validity may therefore diverge temporarily: a volatile slot can still name expert e_s even though macOS is allowed to discard its bytes. The divergence is safe only because a volatile mapping is provisional and cannot directly reach a GPU encoder.

3.4 Relock-before-hit protocol

Table 1 shows the lookup path. Planning runs under the same cache lock that protects identity, LFU/recency data, residency state, and recovery counters. A matching nonvolatile slot is an ordinary hit. A matching volatile slot is first requested as nonvolatile. Metal returns the prior state: retained contents preserve the hit, while an empty return invalidates the mapping and turns the access into a miss. Before overwriting any volatile or empty victim, the runtime also makes that buffer nonvolatile. Only a complete positional read publishes the new expert identity.

Table 1: Reload-correct lookup for requested expert e . Every row executes under the cache lock except the final GPU use.

1	Find the slot s whose published identity equals e , if one exists.
2	If s is volatile, request nonvolatile state and inspect the prior state.
3a	If contents were retained, record the relock and preserve the hit.
3b	If the prior state is empty, clear identity and validity and convert the access to a miss.
4	On a validated hit, update LFU/recency and return the slot.
5	On a miss, select a victim and make its Metal buffer nonvolatile.
6	Read the complete expert blob positionally into the stable buffer pointer.
7	Publish the new identity only after the complete read; then return the exact slot.

3.5 Selective hot and cold residency

At a safe point, occupied slots are ranked first by cumulative use count and then by recency. The configured hot subset remains nonvolatile. Every other occupied nonvolatile slot transitions to volatile. The evaluated policy uses 96 logical slots and a 16-slot hot set per layer. This is not a claim that 16 is universally optimal; it is a frozen policy chosen to combine the identity capacity of the large cache with the physical commitment of a small working set.

The OS remains free to keep volatile pages while memory is plentiful. Under contention it can reclaim them without asking the model runtime to predict the future. ElasticUMA supplies model-specific admission, identity, and exact recovery; macOS supplies global knowledge about competing physical demand.

3.6 GPU-safe transition points

Purgeability cannot change while a live command buffer may read a slot. During chunked prefill, ElasticUMA runs the residency policy after the just-completed layer’s routed tiles and tail command buffer finish. This lets layer i offer its cold pages before layer $i + 1$ expands. During decode, the policy runs after the complete step and output head finish. Both positions reuse synchronization already required for router readback and command completion.

A post-prefill callback is insufficient. In an earlier implementation, every layer reached its 96-slot physical peak before any slot became volatile. The logical policy was correct but the peak-footprint gate failed. Moving the same hot/cold transition into the layer loop changed when physical entitlement was released and passed the successor gate.

3.7 Correctness invariants

The mechanism maintains five invariants:

1. A GPU encoder binds a cached expert only when the slot is known nonvolatile.
2. An empty prior state is never counted or consumed as a hit.
3. A new expert identity is published only after a complete exact read.

4. Purgeability transitions occur only after the last GPU command that may reference the affected resource has completed.
5. Fixed mode executes none of the purgeability path.

Together, the first four reduce OS reclamation to an ordinary exact cache miss. The fifth preserves the baseline and provides a fail-safe mode on unsupported systems.

4 Implementation

The measured implementation is a 629-line change at runtime code commit `ec84269d` over pinned upstream `01f7d5e7`. It is distributed as an Apache-2.0 patch whose complete staged diff is verified before every public build. The executable data path is Swift and Metal; Python is confined to model installation, experiment orchestration, admission, and analysis.

4.1 Cache and runner changes

The patch replaces external slot storage with per-slot Metal-owned shared buffers and adds the three-state residency field. It records volatile transitions, relocks, retained relocks, empty recoveries, invalidated slots, and cumulative recovered bytes. Cache lookup calls the relock path while holding the cache lock. Recovery reuses the existing exact positional reader and does not alter router scores or tensor representation.

The model layer exposes one call that applies the hot/cold policy to all routed expert caches. The prefill runner invokes it after each synchronized layer. The decode runner invokes it after each complete token. This placement prevents front ends from forgetting the transition and keeps model-specific cache details out of the CLI and server.

4.2 Public controls and compatibility

The CLI and loopback server expose three controls:

```
--expert-cache-slots 96
--expert-cache-residency os-managed
--expert-cache-hot-slots 16
```

The public wrapper selects these admitted defaults for catalogued MoE models. It also accepts a completed verified `.gturbo` path when the native runtime already supports that architecture. A data-driven JSON catalog pins model repository, revision, source index hash, manifest identity, routing dimensions, and default cache policy. A new catalog entry does not create native architecture support; new tensor layouts, tokenizers, routers, or kernels still require implementation and model-specific tests.

4.3 Evidence plane

The experiment plane acquires a single-model-worker lock and records a privacy-safe hardware receipt. It checks AC power and low-power mode, samples system and process memory, captures native pressure events, and stores the exact command and binary hashes. Raw runs are immutable. Admission fails closed on timeouts, process overlap, missing token counts, invalid throughput, missing memory samples, guard violations, absent expert telemetry, or required output mismatch.

Physical footprint is sampled with `proc_pid_rusage().ri_phys_footprint`. System snapshots include free percentage, compressor pages, swap use, and page size. Per-run peak compressor and swap values are deltas above the pre-run snapshot. Cumulative recovered bytes count repeated exact reload traffic; they do not measure physical footprint.

4.4 Testing and reconstruction

The upstream package command reports 593 tests: 588 pass, while five existing repacker CLI tests look for an executable named `TurboFieldfareRepack` although the package product is `slipstream-repack`. All new cache, inference, CLI, and server tests pass. The public bootstrap clones the exact upstream commit, applies the bundled patch to the Git index, verifies the full staged diff hash, rejects extra source changes, and builds the repacker, CLI, and API server in release mode.

5 Experimental Methodology

5.1 Research questions

The evaluation asks five questions:

- **RQ1:** Does ElasticUMA outperform the large fixed cache while reducing its physical-memory damage?
- **RQ2:** Does it remain competitive with the strongest small-cache fallback rather than buying memory relief with an unacceptable slowdown?
- **RQ3:** Does the effect persist without a synthetic co-tenant?
- **RQ4:** Does real OS reclamation preserve output under repeated empty-slot recovery?
- **RQ5:** Does the mechanism transfer to a second MoE architecture without model-specific policy tuning?

5.2 Platform and immutable inputs

Table 2: Primary experimental platform and pinned artifacts.

Item	Pinned value	Role
Host	Apple M1 Max, 24-core GPU, 32 GiB, macOS 26.5.2	Primary host
Qwen	Qwen3.6-35B-A3B Q4 at 38740b84... [8]	256 experts, 40 layers
Gemma	Gemma 4 26B-A4B Q4 at 0d77464e... [9]	128 experts, 30 layers
Upstream	slipstream at 01f7d5e7...	Untouched fixed baseline
ElasticUMA	runtime code at ec84269d...	OS-managed candidate
Schedule	One excluded warmup plus five measured rows per arm	Balanced order, AC power

The Qwen packed artifact contains 40 MoE layers with 256 routed experts and eight routed selections per token. The Gemma artifact contains 30 MoE layers with 128 routed experts and eight routed selections. Both use the runtime’s 4-bit packed representation and immutable positional reads. Qwen protocols generate 256 tokens from model-appropriate held-out prompts; Gemma generates 128 because its protocol was frozen separately. Absolute token rates are therefore compared only within a model.

5.3 Baselines and protocols

Every positive protocol has three arms:

- **Upstream-16:** untouched upstream with 16 fixed slots per layer;
- **Upstream-96:** untouched upstream with 96 fixed slots per layer;
- **ElasticUMA-96/16:** 96 logical slots with 16 hot nonvolatile slots and OS-managed cold slots.

V6 runs Qwen with a bounded 4,096 MiB co-tenant. V7 repeats Qwen without a pressure worker. V8 repeats the no-pressure protocol with Gemma and a new prompt. Each protocol uses one excluded warmup and five measured triples in a balanced order. All arms within a protocol share checkpoint bytes, prompt, context, generation limit, greedy decoding, and runtime settings. Candidate and baseline release binaries are hashed in each measured row.

Gate 1 is a separate six-arm Qwen sweep over 16, 24, 32, 48, 64, and 96 fixed slots. It contains 30 measured rows and tests the earlier 15% controller gate. Its data are not reused to tune successor prompts or thresholds.

5.4 Safety and admission

A measured row starts only when the required power and low-power-mode state are valid, no conflicting model process exists, and free memory meets the start threshold. Sampling runs at 0.25 or 0.5 seconds depending on protocol. A row is stopped or rejected on timeout, critical pressure, insufficient free memory, excess swap growth, a malformed runtime result, missing expert telemetry, or an output-parity failure. Failed rows are retained and are not silently retried.

The successor gate requires all 15 planned rows, within-protocol output equality, observed empty recovery, no worse swap or critical pressure, and a material physical-footprint or compressor benefit over upstream-96. Against either baseline, the paired median slowdown and the bootstrap upper bound on slowdown must both be no worse than 10%. The 10% rule is a frozen engineering gate for these experiments, not a universal product definition.

5.5 Metrics and paired analysis

For candidate throughput $t_{c,r}$ and baseline throughput $t_{b,r}$ in matched repetition r , the paired gain is

$$g_r = \frac{t_{c,r}}{t_{b,r}} - 1. \quad (1)$$

We report the median of five paired gains and a deterministic 95% percentile bootstrap confidence interval from 10,000 resamples with seed 104729. Arm summaries report the median decode throughput, hit rate, per-run peak physical footprint, per-run peak compressor growth, and minimum free-memory reading. These intervals describe uncertainty among five fixed matched pairs. They are not population-level significance tests, and we do not aggregate absolute token rates across model-specific prompts.

6 Evaluation

Across V6, V7, and V8, all 45 planned positive rows are admitted. No measured row times out, reaches critical pressure, or grows swap. Each protocol produces one response SHA-256 across all three policies, and every candidate row observes nonzero empty-slot recovery. Paired comparisons use the deterministic 95% bootstrap confidence interval defined in Section 5. We first report the negative precursor, then answer the five research questions.

6.1 Negative precursor: hit rate is not the objective

Figure 1 shows the fixed-cache sweep. Hit rate rises monotonically from 48.98% at 16 slots to 90.13% at 96 slots. Decode throughput moves in the opposite direction, while minimum free memory falls from 51% to 32% and peak compressor growth rises from 714.8 MiB to 5,830.0 MiB. The matched 16-to-96 slowdowns are 10.72%, 10.84%, 10.54%, 9.45%, and 8.94%. Their paired median is 10.54%, with a bootstrap interval of 8.94–10.84%.

The tradeoff is stable but below the preregistered 15% controller gate. Gate 1 is therefore rejected. This matters to the positive result: ElasticUMA is not a post-hoc cache size selected from the sweep. It changes memory ownership and validity semantics, and its tests use new prompts and experiment identifiers.

6.2 RQ1–RQ3: Qwen throughput and physical pressure

Table 3: Qwen3.6 policy results. Values are medians of five admitted measured runs per arm. Footprint is peak `ri_phys_footprint`; compressor is the per-run peak delta above the pre-run snapshot.

Protocol	Policy	tok/s	Hit	Footprint	Compressor	Min free
4 GiB co-tenant	Upstream-16	13.025	47.91%	1.374 GiB	0.675 GiB	57%
	Upstream-96	12.116	90.81%	6.705 GiB	5.890 GiB	38%
	ElasticUMA-96/16	14.136	86.83%	4.584 GiB	0.840 GiB	44%
No pressure worker	Upstream-16	14.520	49.82%	1.386 GiB	0.000 GiB	57%
	Upstream-96	14.293	90.58%	6.721 GiB	5.991 GiB	38%
	ElasticUMA-96/16	16.807	87.50%	4.594 GiB	0.700 GiB	44%

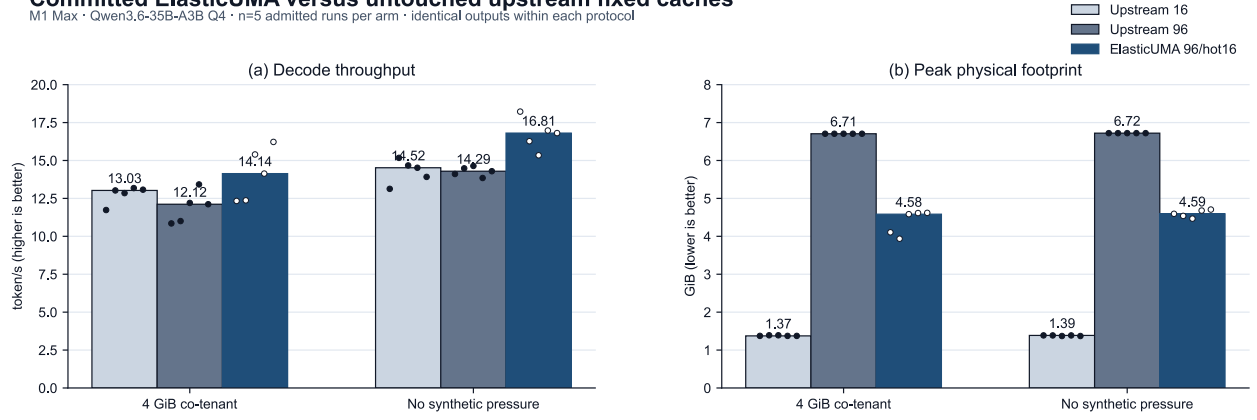
Under the 4 GiB co-tenant, ElasticUMA gains 7.25% paired-median throughput over upstream-16 and 13.67% over upstream-96. The interval against upstream-16 spans -5.02% to +24.08%; against upstream-96 it is +5.33% to +33.91%. The candidate wins nine of ten pressure-backed comparisons and its only loss is 5.02%, inside the frozen fallback gate. It retains 38.92 percentage points more hit rate than upstream-16.

Against upstream-96, peak physical footprint falls from 6.705 to 4.584 GiB, a 31.64% reduction. Peak compressor growth falls from 5.890 to 0.840 GiB, a 5.05 GiB benefit. ElasticUMA is not the smallest-memory policy: upstream-16 has a 1.374 GiB footprint. The candidate dominates upstream-96 in the measured throughput-footprint plane. Against upstream-16 it trades a larger footprint for higher median throughput and 38.92 percentage points more locality.

Without a synthetic pressure worker, the same policy gains 16.91% over upstream-16 (interval 4.56–38.79%) and 17.59% over upstream-96 (interval 4.79–29.15%). It wins all ten matched comparisons. Fixed-96 footprint falls 31.65%, and compressor growth falls by 5.29 GiB. The effect therefore does not require the controlled co-tenant, although the absolute rates differ because the host begins from a different memory state.

Committed ElasticUMA versus untouched upstream fixed caches

M1 Max · Qwen3.6-35B-A3B Q4 · n=5 admitted runs per arm · identical outputs within each protocol

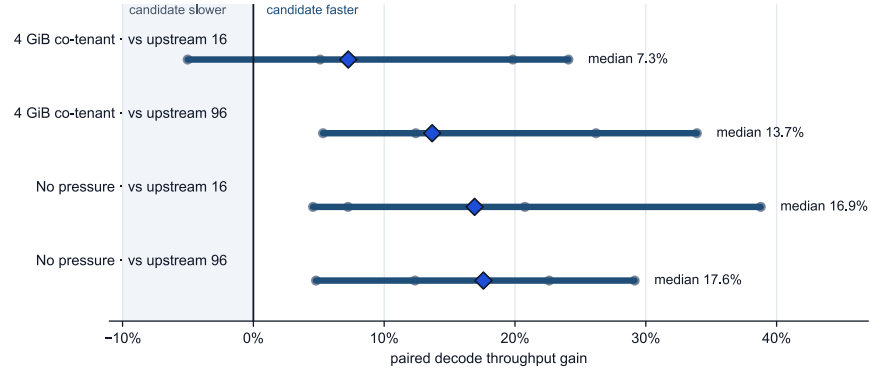


Bars show medians; dots show all measured runs. Source: admitted V6/V7 receipts.

Figure 3: Qwen3.6 policy comparison. With the co-tenant, median throughput is 14.136 token/s for ElasticUMA versus 13.025 and 12.116 token/s for upstream-16 and upstream-96. Without it, the values are 16.807, 14.520, and 14.293 token/s. Fixed-96 footprint falls by about one third under both protocols. Bars are arm medians; points are all measured runs.

Paired decode throughput gain by protocol and baseline

Candidate gain = (ElasticUMA - upstream) / upstream; five same-repetition pairs per row



Dots: measured pairs · diamond: median · navy interval: deterministic bootstrap 95% interval.

Figure 4: Matched Qwen decode gains against upstream-16 and upstream-96. Paired medians are 7.25% and 13.67% with the co-tenant and 16.91% and 17.59% without it. Horizontal intervals are deterministic bootstrap 95% intervals from five matched repetitions; the shaded region marks slowdowns.

6.3 RQ4: recovery is exercised, not inferred

Every Qwen candidate row records volatile transitions, relocks, and a nonzero empty-recovery count. Cumulative recovered bytes range from 9.090 to 10.758 GiB per pressure-backed run and from 7.907 to 10.331 GiB per no-pressure run. These totals exceed instantaneous footprint because a cold slot can be reclaimed and reloaded repeatedly.

All policies within V6 produce one identical response hash; V7 produces a second within-protocol hash. These matching hashes demonstrated response-text equality under greedy decoding despite real empty recovery. The current upstream CLI does not expose generated token IDs, so we do not claim token-ID equality. The evidence is stronger than a no-reclamation test but narrower than a full numerical trace comparison.

6.4 RQ5: transfer to Gemma 4

Table 4: Gemma 4 second-architecture validation without a pressure worker. Values are medians of five admitted measured runs per arm.

Policy	tok/s	Hit	Footprint	Compressor	Min free
Upstream-16	16.394	65.52%	1.986 GiB	0.000 GiB	69%
Upstream-96	18.742	98.40%	8.538 GiB	5.114 GiB	50%
ElasticUMA-96/16	23.820	95.79%	5.527 GiB	1.073 GiB	55%

Gemma changes layer count, expert count, tokenizer, routing, and per-slot size. To test architecture transfer without tuning, we preserve the Qwen 96/16 residency policy. Against the pinned upstream-16 and upstream-96 policies [7], ElasticUMA wins every matched comparison. Paired-median gains versus upstream-16 and upstream-96 are 45.30% (interval 23.45–51.44%) and 26.01% (interval 11.88–28.55%), respectively. Relative to fixed-96, peak footprint falls 35.26%, from 8.538 to 5.527 GiB; compressor growth falls from 5.114 to 1.073 GiB. Hit rate remains 30.28 percentage points above the small cache. All candidate rows exercise exact empty recovery (2.421–2.868 GiB of cumulative reload traffic), and no row grows swap.

No-pressure result across two MoE architectures

M1 Max · n=5 admitted runs per arm · model-specific unseen prompts · exact outputs

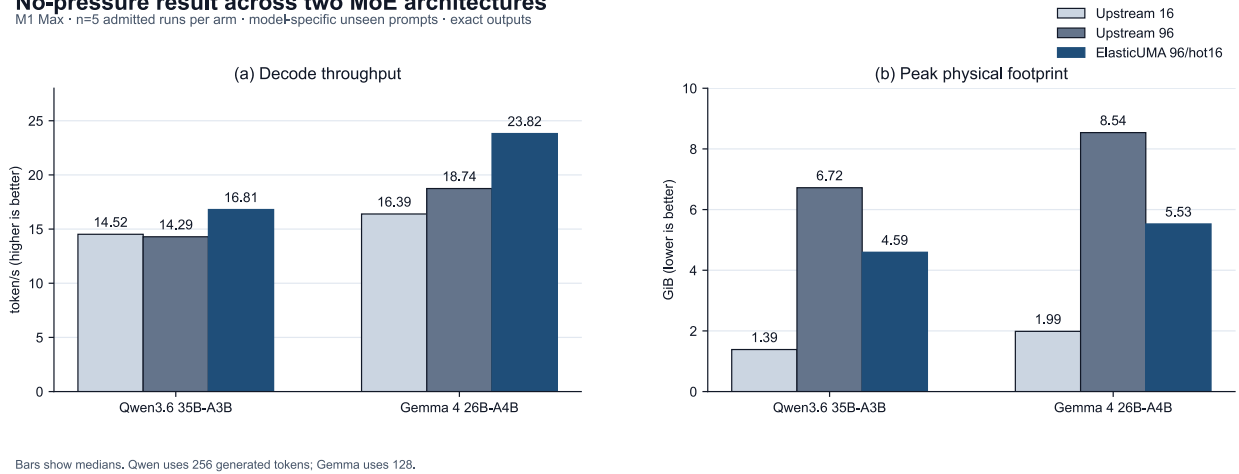


Figure 5: No-pressure results across two MoE architectures. Absolute token rates are not a cross-model benchmark because prompts and generation lengths differ. Within each model, ElasticUMA has the highest median decode rate and a materially lower footprint than upstream-96.

6.5 Rejected alternatives and mechanism ablation

Table 5 records the measured failure criterion for each rejected mechanism. This is a mechanism-screening record, not a quantitative aggregate across incompatible prototypes.

Table 5: Alternatives rejected before the final per-slot, layerwise design.

Alternative	Observed failure
F_NOCACHE	Advice changed file-cache treatment but did not release the anonymous slot allocation.
Direct mmap	Reduced measured model RSS from 3.38 to 1.92 GiB in a probe, but was approximately 24 times slower than the cached positional-read path.
Raw/compressed MTLIO	Direct-storage variants missed frozen latency gates; lossless CPU decode reached only 0.46–2.85 GiB/s.
One arena per layer	Coarse purgeability collapsed Qwen effective hit rate to 1.3% and roughly halved throughput.
Post-prefill-only policy	Correctly marked cold slots but ran after the full cache had already materialized, failing the footprint gate.

Only Metal-owned per-expert buffers, selective hot/cold state, and transitions inside the synchronized layer loop satisfy both validity and footprint gates.

6.6 Dense Qwen3.8 diagnostic

We also run one canonical Qwen3.8-27B Q4 dense checkpoint with MLX [10]. One warmup and one 128-token measured row complete at 11.36 token/s. The sampled peak physical footprint is 17.96 GiB; MLX reports 18.60 GiB peak memory. Minimum free memory reaches 20%, and swap grows by 151 MiB. A later row reaches 19% free memory and is terminated by the frozen guard. Because the planned five measured rows do not complete, this diagnostic is not admitted. We do not use it as a speed, quality, or dense-versus-MoE comparison.

7 Discussion

7.1 Why fewer physically entitled pages can be faster

ElasticUMA does not make SSD reads free and does not eliminate expert misses. Its Qwen hit rate remains near 87%, below fixed-96 near 91% and far above fixed-16 near 49%. The improvement comes from a different system balance. The large logical cache still captures reuse when cold pages survive, while the OS can reclaim cold pages instead of compressing them or displacing more useful file-backed state. The fixed-96 baseline shows that maximizing hit rate can reduce service rate once its physical claim harms the shared pool.

This interpretation is supported by three observations. First, physical footprint and compressor growth fall materially relative to fixed-96. Second, the candidate leads in no-pressure experiments as well as under the controlled co-tenant, so the result is not solely an artifact of one pressure process. Third, repeated empty recovery occurs in every candidate row without output divergence. The mechanism is therefore exercised in the measured path rather than acting as an unused safety annotation.

7.2 The OS as eviction partner, not model scheduler

A recent kernel-cache study finds that ordinary OS replacement can be a strong expert tier when the kernel sees the complete physical workload [11]. ElasticUMA reaches a related design point without delegating model identity or exact publication to the kernel. The runtime decides which pages are eligible, retains hot experts, validates slot contents, and reloads exact weights. macOS decides whether eligible pages should remain physical given global demand. This division avoids a runtime-only estimate of desktop competition while preserving model-specific correctness.

The design is not a complete adaptive memory controller. It uses a fixed logical/hot split and lets the OS react within the cold set. That simplicity is a strength of the current evidence: there is no learned predictor, online exploration, or second model configuration that can perturb the state being measured. Future work can adapt the hot-set size, but it must beat this non-destructive baseline and account for transition cost.

7.3 Relationship to discrete-GPU edge serving

FreeToken treats CPU memory, GPU memory, PCIe bandwidth, and CPU compute as heterogeneous resources and continuously maps work across them [1]. ElasticUMA adopts its challenge-driven systems discipline but not its CUDA data plane. On Apple Silicon there is no expert copy across a host/device boundary. The relevant elasticity is whether cold shared pages remain physically valid. CPU execution, bandwidth-adaptive miss splitting, semantic checkpoints, and agent lifecycle management remain complementary.

7.4 Implications for model support

The mechanism transfers from Qwen’s 256-expert, 40-layer layout to Gemma’s 128-expert, 30-layer layout without policy tuning. That supports an architecture-independent cache-validity abstraction, not arbitrary-model compatibility. A new model still needs a correct tokenizer, packed manifest, tensor mapping, router semantics, quantization decoder, and Metal kernels. ElasticUMA’s public catalog makes checkpoint metadata data-driven, while the native runtime remains the correctness boundary.

7.5 Practical deployment

The public CLI defaults to OS-managed 96/16 residency for admitted profiles and offers one-shot generation plus a loopback OpenAI/Anthropic-shaped server. It reuses one canonical model store, pins revisions, serializes downloads, and refuses a transfer that would violate its disk reserve. These product surfaces are engineering contributions for reproducibility and adoption; they are not part of the measured algorithmic claim.

8 Related Work

8.1 Edge MoE serving and expert caching

Edge MoE systems exploit sparsity through different resource assignments. FreeToken co-designs residency, prefill streaming, CPU-GPU miss execution, and elastic VRAM management for heterogeneous CUDA machines [1], whereas KTransformers moves selected MoE work to CPUs and optimizes the hybrid path [2]. MoE-Infinity instead uses request-level activation traces to guide prefetching and caching on personal machines [5]. Taken together, these systems establish sparse local execution, but their main physical boundary is host versus device memory; they do not control the lifetime of explicit Metal slot pages inside one shared Apple pool.

MawForge stores full MoE weights on disk and materializes routed experts into a bounded cache on unified-memory machines [6], while slipstream supplies native Metal execution, packed storage, positional reads, and the fixed-cache baseline used here [7]. MawForge’s cache-size result motivates an end-to-end memory view; slipstream isolates the residency mechanism under otherwise identical execution. Their shared limitation is that a stable expert slot has no validity protocol after the OS may discard its physical contents. ElasticUMA does not claim their storage, Metal, or LFU mechanisms as new.

8.2 Representation and transfer reduction

SliceMoE expands effective cache capacity through bit-sliced reconstruction [12], whereas ZipMoE combines lossless expert compression with cache-affinity scheduling [13]. ExactMoE focuses on exact low-bit slots and fused kernels [14]. Despite changing different parts of the representation and transfer path, all three leave a separate residency question: a compressed, sliced, or fused cold slot still needs a validity rule if its physical pages may disappear. Overall, representation savings and residency validity are complementary rather than competing solutions.

8.3 Apple execution engines

BaseRT improves native Metal kernels and dispatch [15], while FusionML identifies the eager boundary that enables CPU-GPU prefill co-execution and the bandwidth limit that prevents a similar decode benefit [16]. NPUMoE maps static work to the Apple Neural Engine but keeps CPU/GPU fallbacks for dynamic operations [17]. Collectively, these systems optimize where or how computation runs. ElasticUMA changes neither placement nor kernel math; it defines a reload-correct, OS-discardable lifetime around the existing Metal path.

8.4 Memory-aware runtimes and kernel ownership

PagedAttention separates logical KV state from physical allocation to reduce fragmentation [18], while MemSpec coordinates draft selection with resident memory on edge devices [19]. By contrast, the kernel-managed expert-cache study delegates replacement to the page cache and finds that it can rival model-specific frequency policies under equal limits [11]. ElasticUMA follows the shared separation principle but adds Metal-resource validity, GPU-safe transitions, and exact reload for weights that retain stable logical slots. Overall, these runtimes separate logical state from physical placement at different layers; routed Metal weights add a validity transition that they do not cover.

8.5 Novelty boundary

No individual component is claimed in isolation: expert caching, storage-backed weights, Metal inference, LFU ranking, OS page replacement, and purgeable graphics resources all predate this work. The contribution is their reload-correct composition at per-expert granularity and real inference phase boundaries, plus evidence that the composition improves the measured throughput-footprint frontier against untouched fixed policies.

Across these categories, a specific gap remains among the cited systems: none evaluates per-expert Metal resources that macOS may reclaim while their logical cache identities survive, with relock-before-hit validation and exact recovery at real inference boundaries. ElasticUMA addresses that gap without claiming the surrounding storage, caching, or Metal mechanisms as new.

9 Limitations, Validity, and Ethics

Hardware breadth. All positive evidence comes from one 24-GPU-core M1 Max with 32 GiB memory. Qwen and Gemma provide architectural diversity on the same host, not cross-generation evidence. M2 through later Apple generations may expose different memory, GPU, and purgeability behavior.

Workload breadth. Experiments use batch size one, short held-out prompts, one 4,096-token context limit,

and one controlled 4 GiB co-tenant. Real desktop traces, longer contexts, concurrent requests, multimodal inputs, and tool-heavy sessions remain unevaluated. Qwen and Gemma use different prompts and generation lengths, so cross-model absolute token rates are not comparable.

Measurement. macOS pressure events and compressor statistics are system-wide and coarse. Physical footprint is process-specific and is the primary memory metric, but it does not identify every causal path. Energy and GPU power are unmeasured. Five matched repetitions support descriptive bootstrap intervals, not population-level inference.

Exactness. Within-protocol response text is identical under greedy decoding, and every candidate row exercises empty recovery. The pinned CLI does not expose token IDs or per-layer numerical traces. We therefore avoid claims of bitwise intermediate equality or model-quality improvement.

Baselines. The controlled baselines are the untouched fixed policies of the same native runtime. This isolates the mechanism but does not establish superiority over every version of MLX, llama.cpp, BaseRT, or another Mac engine. Cross-runtime comparisons require matched model bytes, quantization, prompts, context, output semantics, and memory accounting.

Artifact and privacy. The public artifact excludes model weights, credentials, serial numbers, UUIDs, personal paths, and unrelated process commands. Model checkpoints retain their own licenses and acceptable-use terms. The loopback API has no authentication or TLS and must not be exposed to a LAN or WAN. AI assistance is disclosed rather than treated as an anonymous author; the named author remains responsible for verification and submission policy.

10 Conclusion

Our experiments demonstrate that a fixed expert cache equates logical capacity with permanent physical entitlement. That assumption is costly on Apple unified memory, where model slots, file-backed pages, GPU work, applications, compression, and swap share one pool. ElasticUMA separates the two. It preserves a large logical cache, offers cold Metal-owned pages to macOS at GPU-safe boundaries, and converts every discarded provisional hit into an exact reload.

On one M1 Max, the committed mechanism improves median decode throughput over both small and large fixed upstream policies for Qwen3.6 and Gemma 4 while reducing the large cache’s peak physical footprint by roughly one third. The contribution is the demonstrated separation of cache identity from physical entitlement with exact recovery under real OS reclamation. The negative precursor, failed alternatives, and incomplete dense control remain part of the evidence record. The result is a systems mechanism scoped to the tested host and workloads. The practical implication is that an Apple runtime can retain cache identity without insisting that all cold pages remain physical. The next decisive steps are independent cross-generation reproduction, energy measurement, longer and concurrent workloads, and comparison with strong external Mac runtimes under matched model semantics.

Artifact Availability

The source, exact upstream patch, experiment harness, admitted receipts, and analysis programs are available in the public repository linked on the first page. The tagged artifact archive includes internal SHA-256 checksums and the raw evidence used to derive every reported table and quantitative figure. Model weights, credentials, machine identifiers, and local paths are excluded. Model checkpoints must be obtained from their original publishers under their own licenses.

Artifact layer	Public location
Runtime mechanism	runtime/patches/ and scripts/bootstrap_candidate_runtime.sh
Protocols and admission	configs/, src/elasticuma/admission.py, and docs/METHODOLOGY.md
Measured evidence	artifacts/releases/ with raw and admitted JSON receipts
Analysis and figures	scripts/, docs/RESULTS.md, and paper/figures/
Paper source	paper/latex/, including the bibliography and vector figures

The reconstruction sequence and hardware-reporting fields are documented in docs/REPRODUCIBILITY.md. The repository deliberately keeps negative results and failed alternatives in the same evidence lineage as admitted results; reproductions that disagree with the reported measurements should be retained rather than tuned away.

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